NATIONAL UNIVERSITY OF SCIENCES & TECHNOLOGY
School of Electrical Engineering and Computer Science (SEECS)

CS-477 Computer Vision

Semester Project | SEECS CV Face-Off 2026

PROJECT ARCHITECTURE REPORT

Driver Drowsiness Monitoring System

SG-4 · Temporal Behaviour Analysis

Converting frame-level cues into persistent temporal indicators of drowsiness

First Evaluation | Group G1-D2 (Gp 1)

SUBMITTED BY

Saman Iqbal
CMS ID: 457622

Syeda Musfirah Hamid
CMS ID: 410933

September 2026

Module Overview

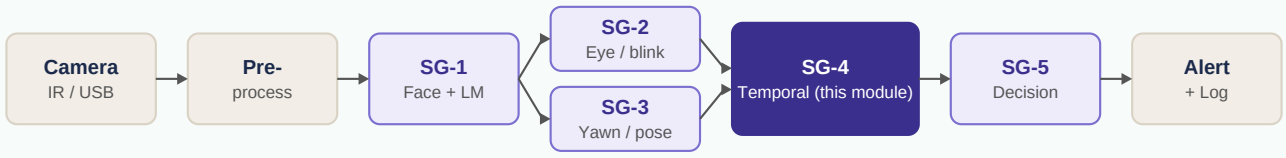
Problem Statement

Drowsiness is inherently temporal: a single closed-eye frame may be an ordinary blink, while the same closure held for half a second is a microsleep. SG-4 sits between the per-frame cue extractors (SG-2, SG-3) and the decision logic (SG-5). It converts frame-level observations into persistent temporal indicators by modelling the **duration, frequency and temporal persistence** of each cue, rather than relying on isolated frame classifications. Its output is the evidence on which every drowsiness decision is made.

Objectives

- Buffer per-frame cues from SG-1/2/3 in a 30–60 s sliding window aligned on a single timestamp clock.
- Compute established fatigue indicators: PERCLOS (P80), blink rate and duration, microsleeps, yawn and nod frequency.
- Normalise indicators against a per-driver baseline and report their trend, so slow deterioration is visible early.
- Deliver a compact indicator vector to SG-5 at 10 Hz with negligible load on the Jetson Orin Nano.

Position in System Architecture



SG-6: Jetson Orin Nano 8GB runtime (SG-4 runs as a lightweight CPU thread between cue extraction and decision)

Temporal Indicators Produced

Indicator	Definition (over sliding window)	Why it signals drowsiness
PERCLOS (P80)	Share of time eyes are at least 80% closed	Most validated single fatigue measure
Blink rate	Blinks per minute from SG-2 blink events	Rises, then becomes irregular, with fatigue
Blink duration	Mean closure time per blink; count of long blinks (> 400 ms)	Blinks slow down before sleep onset
Microsleeps	Continuous closures of 0.5 s or longer; longest closure	Direct sign of lost vigilance
Yawn frequency	Yawns per minute and mean duration from SG-3	Early, pre-closure fatigue cue
Nod / head drop	Nod events and time with head pitched down	Late-stage cue, near loss of control
Trend & validity	Slope vs driver baseline; share of valid frames	Separates worsening from stable; flags weak data

How It Works

Each cue record from SG-1/2/3 is time-aligned and validated; frames where the face is lost are marked invalid rather than closed, so looking away is never mistaken for sleep. Records enter a ring buffer holding the last 30–60 s. Running sums update PERCLOS and event counts in constant time, while a small event tracker turns consecutive closed frames into blinks, long blinks and microsleeps. A 60 s calibration phase at the start of each drive sets the driver's personal baseline, and every indicator is reported relative to it together with its short-term trend. The resulting vector is published to SG-5 through a frozen interface, so SG-4 can be developed and tested against recorded or mock cue streams before integration.

Evaluation & Targets

- **Accuracy:** PERCLOS within 5 points of manual annotation; microsleep recall ≥ 90%; blink-rate error under 2 blinks/min.
- **Timeliness:** indicator lag ≤ 1 s after a microsleep ends; update cost under 2 ms on the Jetson CPU.
- **Ablation:** compare temporal indicators against single-frame classification to show the reduction in false alerts per hour.
- **Data:** NTHU-DDD and YawDD video sequences, plus team-recorded clips with annotated blinks and microsleeps.

SG-4 Module Flowchart

Internal pipeline of the temporal analysis module, from incoming per-frame cues to the indicator vector sent to SG-5. Purple = SG-4 processing stage, grey = input/output interface, dashed boundary = SG-4 module.

